CH365 Chemical Engineering Thermodynamics

Lesson 29
Residual Properties II

Agenda
Review
Finish PS10
Capstone

Block 6 Look-Ahead (CDP)

Real Gas Properties

- Residual Properties
- M = V, U, H, S, or G

$$M^R \equiv M - M^{ig} \qquad \text{(Eq. 6.41)} \qquad M \equiv M^{ig} + M^R$$

Ideal gas (ig) follows ideal gas law

Real Solution Properties (Liquids)

- Excess Properties
- $M^E = V^E, U^E, H^E, S^E$, or G^E

$$M^E \equiv M - M^{id} \qquad \text{(Eq. 6.41)} \qquad M \equiv M^{id} + M^E$$

$$G^E \equiv G - G^{id} \qquad H^E \equiv H - H^{id} \qquad S^E \equiv S - S^{id}$$

Ideal solution (id) follows Raoult's law $\left(y_i P = x_i P_i^{sat}\right)$.

G^E is related to the activity coefficients

From Gibbs-Duhem:

$$\frac{G^E}{RT} = \sum_i x_i \ln \gamma_i \qquad \text{(Eq. 13.10)}$$

Margules (1-constant)

$$\log \gamma_1 = A x_2^2 \qquad \log \gamma_2 = A x_1^2$$

- The constant A is known from fitting experimental data.
- These are the so-called "binary interaction parameters" in CHEMCAD and in the *capstone design problem*.

Residual Properties from Cubic EOS

Numerical Recipe from Lesson 28 (red equations) **(WPR3)**

Important: α is written as a function of x where x replaces the reduced temperature T_r.

This is a purely computational issue that arises from the order of operations in the algorithm.

The derivative of q that appears in H^R and S^R must be determined before T_r is assigned a value in the Mathematica kernel.

$$\alpha = \alpha(\omega, x)$$ instead of $\alpha = \alpha(w, Tr)$ (α formula is from Table 3.1)

$$q = q(x) = \frac{\Psi \alpha(x)}{\Omega \cdot x}$$ (eq. 3.51)

$$\beta = \Omega \frac{P_r}{T_r}$$ (eq. 3.50)

$$Z = 1 + \beta - q\beta \frac{Z - \beta}{(Z + \varepsilon\beta)(Z + \sigma\beta)}$$ (eq. 3.48)

Generic cubic equation of state in Z-Form.

RK, SRK, PR: $$I = \frac{1}{\sigma - \varepsilon} \ln\left(\frac{Z + \sigma\beta}{Z + \varepsilon\beta}\right) \quad \varepsilon \neq \sigma$$ (eq. 13.72)

vdW: $$I = \frac{\beta}{Z} \quad \varepsilon = \sigma$$

$$\frac{H^R}{RT} = Z - 1 + T_r \left(\frac{dq}{dT_r}\right) \cdot I$$ (page 497)

$$\frac{S^R}{R} = \ln(Z - \beta) + \left(q + T_r \frac{dq}{dT_r}\right) \cdot I$$ (page 497)

Review of Lessons 27 and 28

These equations originated from equation 6.37:

$$d\left(\frac{G}{RT}\right)=\frac{V}{RT}dP-\frac{H}{RT^2}dT \qquad \text{(Eq. 6.37)}$$

Equation 6.37 arose from the total differential of G as a function of T and P combined with the fundamental property relation for G and the definition of G (G=H-TS) (See L27 Slide 21).

Equation 6.37 was then re-written as a residual to give 6.42 (See L28 slide 5):

$$d\left(\frac{G^R}{RT}\right)=\frac{V^R}{RT}dP-\frac{H^R}{RT^2}dT \qquad \text{(Eq. 6.42)}$$

We then equated the coefficients to those of the total derivative of G^R/RT, integrated V^R/RT to obtain G^R, applied the definition of V^R to bring in Z, and took the derivative with respect to T to give H^R (See L28 slide 6).

S^R followed forward from the definition of G^R ($G^R=H^R-S^R$) (See L28 slide 7).

These Appear in Lesson 28 slide 7.

$$Z=1+\beta-q\beta\frac{Z-\beta}{(Z+\varepsilon\beta)(Z+\sigma\beta)}$$

$$V^R=\frac{RT}{P}(Z-1) \qquad \text{(Eq. 6.40)}$$

$$\frac{H^R}{RT}=-T\int_0^P\left(\frac{\partial Z}{\partial T}\right)_P\frac{dP}{P} \qquad \text{(Eq. 6.46)}$$

$$\frac{S^R}{R}=-T\int_0^P\left(\frac{\partial Z}{\partial T}\right)_P\frac{dP}{P}-\int_0^P(Z-1)\frac{dP}{P} \qquad \text{(Eq. 6.48)}$$

$$\frac{G^R}{RT}=\int_0^P(Z-1)\frac{dP}{P} \qquad \text{(Eq. 6.49)}$$

With a modern tools we should be able to integrate these directly, verifying the theory.

Questions?